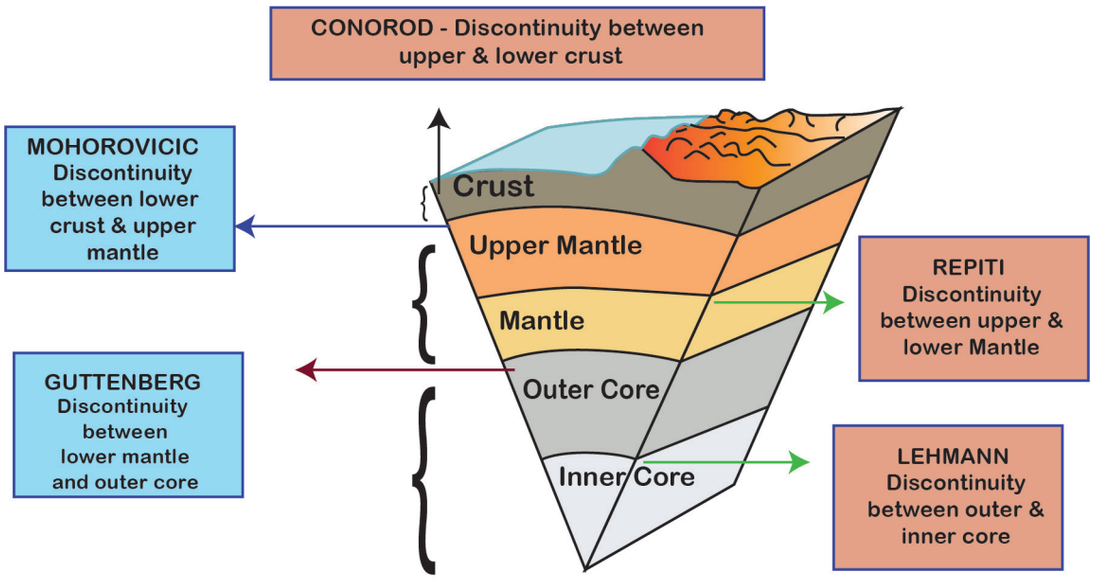


Figure 3.4 Layers of the Earth

3.2 Interior of the Earth

The interior of the earth is composed of many minerals both in the solid and liquid state. The temperature in general increases at the rate of 1° C for every 32 metres towards the earth's interior.

Look at the figure.3.4 the layers of the earth. Earth's interior can be divided into the crust, upper mantle, lower mantle, outer core, and inner core.

3.2.1 The Crust

The crust is further divided into upper crust (continental crust), composed of silica and aluminum (sial) and the lower crust (oceanic crust) made up of silica and magnesium (sima). The boundary between the upper crust and the lower crust is termed as 'Conorod boundary'. The thickness of the crust varies from oceanic areas to continental areas. Oceanic crust is thinner when compared to the continental crust. The mean thickness of oceanic crust is 5 km while the continental crust is around 30 km. The continental crust is thicker in the areas of major mountain systems. It is as much as 70 km thick in the Himalayan region. The density of the crust is less than 2.7 g/cm^3.

3.2.2 The mantle

The mantle is composed of silica, magnesium and iron. It lies between the lower crust and the outer core. It extends for about 2,900 km. It is divided into upper mantle and lower mantle. The mantle generally is in a solid state. The upper part of the mantle is called asthenosphere. The word *Asthen* in Greek means weak. It extends up to 400 km and it is the main source of magma. The Mohorovicic is the boundary which divides the lower crust and the upper mantle. The density of the mantle is 3.9 g/cm^3.

3.2.3 The core

The core forms the centre of the earth. Its density is 13.0 g/cm^3. Its temperature is about 5500° C to 6000° C. The core has two parts namely the outer core and the inner core. The boundary between the lower mantle and the outer core is called Guttenberg margin.

The outer core and inner core are separated by **Lehmann boundary**. The

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